One-To-One Functions

A function $f: A \to B$ associates a single $b \in B$ to each $a \in A$, but the same $b \in B$ may be associated to several values of A.

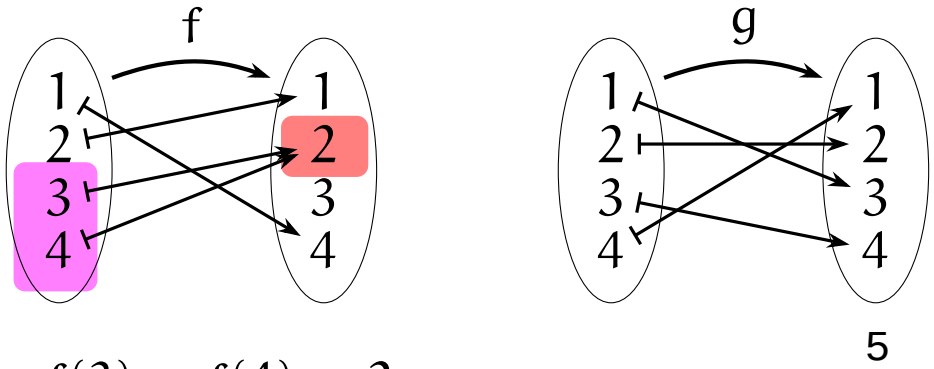
For example, for $f(x) = x^3 - x$ we have $f(-1)=0$ and $f(0)=0$ and $f(1) = 0$.

If f does *not* assign the same value to different inputs, it is called ***one-to-one*** (1-1), or ***injective***:

$$\forall\, a, a' \in A \left(a \neq a' \;\Rightarrow\; f(a) \neq f(a')\right).$$

For all a ∈ A, a' ∈ A, if a ≠ a' then f(a) ≠ f(a').

Example: Consider the following functions:



f is not injective, as $f(3) = f(4) = 2$.

However, g is injective.

One-To-One Functions

Exercise: Which of the following functions are one-to-one?

1. ~~The function score: Class $\to$ Marks.~~

2. The function $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$. `injective!`

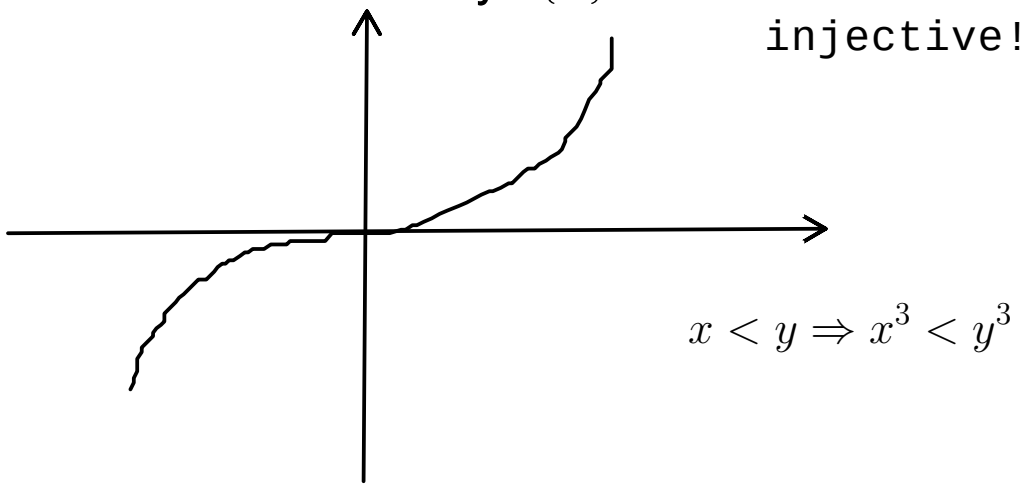
$$n < m \Rightarrow n^2 < m^2$$

3. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$. `not injective!`

`f(-1) = f(1)`

4. The function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$.

`injective!`



$$x < y \Rightarrow x^3 < y^3$$

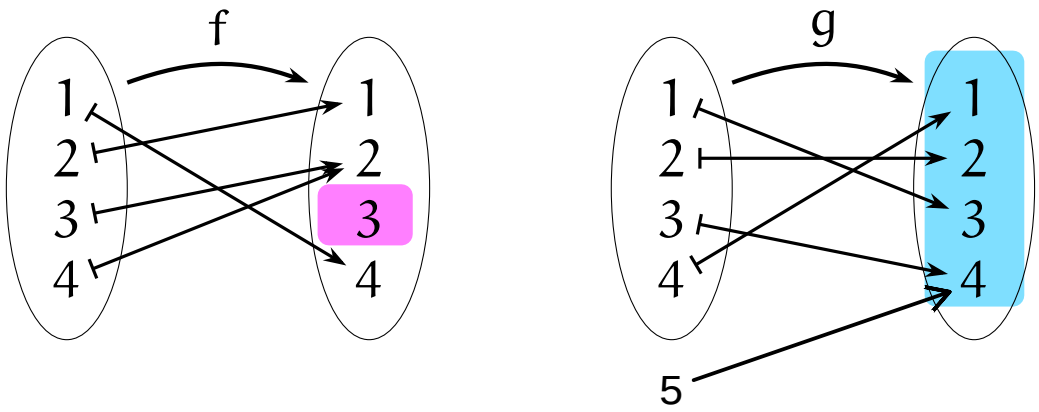
Onto Functions

A function $f: A \to B$ is ***onto***, or ***surjective***, if, and only if, every value in its codomain is produced by the function; that is, its range equals its codomain:

$$\forall b \in B \; \exists a \in A \; \big(f(a) = b\big).$$

For all b ∈ B there exists a ∈ A such that f(a) = b.

Example: Consider the following functions:



f is not surjective, as the value 3 is not produced by the function.

However, g is surjective.

Onto Functions

Exercise: Which of the following functions are onto?

1. ~~The function score: Class $\to$ Marks.~~

2. The function $f: \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$.

   `{0, 1, 4, 9, 16, 25, ...}`

   `This function is not surjective, e.g. 2 is not the square of any natural number.`

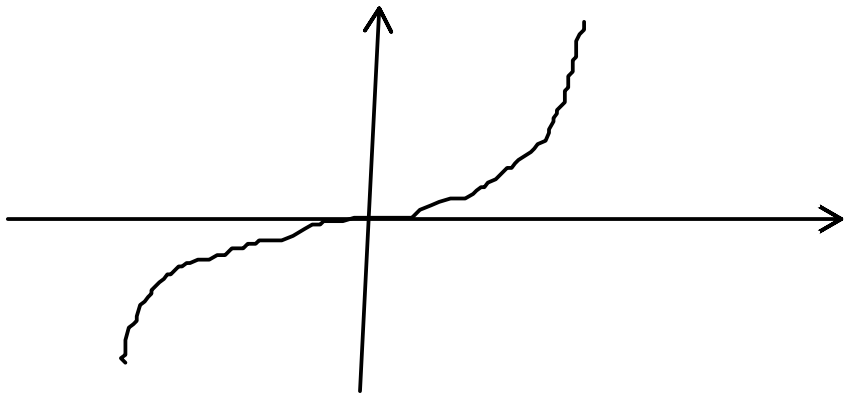
3. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

   $x^2 \geq 0$ for all $x \in \mathbb{R}$

   `This function is not surjective. For example, -1 is not the square of any real number!`

4. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$.

   `This function is surjective!`

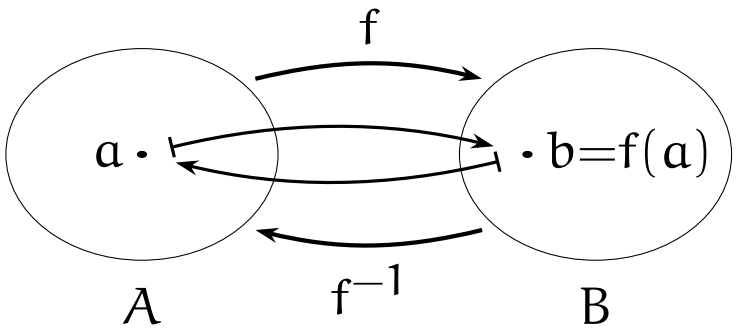


Bijective Functions

A function $f\colon A \to B$ is ***bijective***, or a ***bijection***, if, and only if, it is both one-to-one and onto.

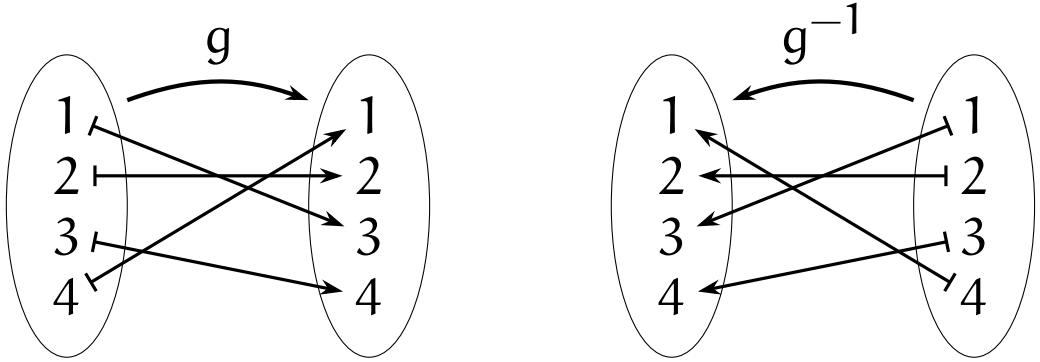
The arrows of a bijection can be reversed to give the ***inverse function*** $f^{-1}\colon B \to A$ that assigns to each $b \in B$ the unique $a \in A$ such that $f(a)=b$:

$$f^{-1}(b) = a \text{ if, and only if, } f(a)=b$$

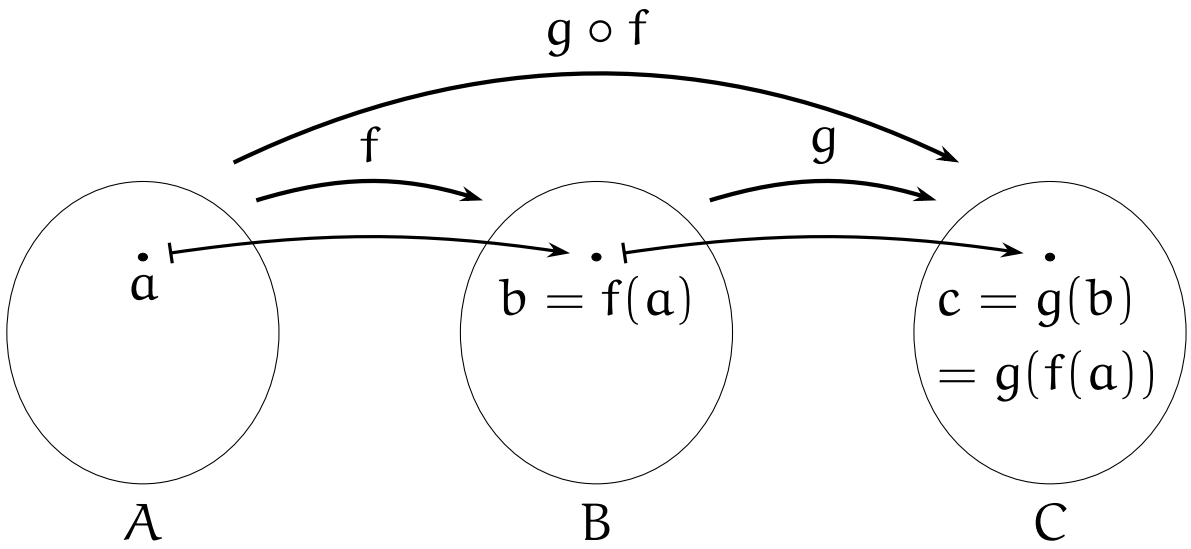


$f^{-1}\colon B \to A$ is also a bijection, and $(f^{-1})^{-1} = f$.

Example:



Properties of Composed Functions



- If $f : A \to B$ and $g : B \to C$ are both one-to-one,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both onto,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both bijective,

  then so is $g \circ f : A \to C$.

Prop. Let A, B be finite sets.

If f is a surjective function from A to B, then $|A| \geq |B|$

If f is an injective function from A to B, then $|A| \leq |B|$

If f is a bijective function from A to B, then $|A| = |B|$

Conversely, if the cardinality of A is greater than or equal to the cardinality of B, then there exists some surjective function from A to B and some injective function from B to A.

If A and B have the same cardinality, then there exists some bijection between A and B.

Cultural Remark.

The above proposition yields a definition for comparing the cardinality of infinite sets: Two not necessarily finite sets have the same cardinality if and only if there exists a bijective function between them.

Example: the natural numbers, the integers, the rational numbers all have the same cardinality.

But the real numbers have a strictly greater cardinality than the integers.

The powerset of a set has a strictly greater cardinality than the set itself.

Def.

Let $f: A \to B$ $g: B \to A$

We say that g is a left inverse of f if

$$g \circ f(x) = x$$

for all x in A.

We say that g is a right inverse of f if

$$f \circ g(x) = x$$

for all x in B.

g is an inverse of f if and only if it is both a left inverse and a right inverse.

Fact.

A function f is injective if and only if it has a left inverse.

A function f is surjective if and only if it has a right inverse.

A function f is bijective if and only if it has an inverse.